Ashvin Anilkumar

ashvinanilkumarsh@gmail.com | linkedin.com/in/ashvin-anilkumar | github.com/ashvin-a | +1(608)-395-4238

Summary

Graduate student specializing in Robotics and AI, passionate about advancing embodied intelligence through robot learning and perception. Experienced in developing autonomous systems that integrate reinforcement learning with real-world robotic control. Strong foundation in machine learning, with a research focus on scalable perception, learning-based control in complex human environments and building high performance software for multi-modal learning.

Education

University of Wisconsin-Madison, Master of Science in Electrical and Computer Engineering — Aug 2025 - Dec 2026

- *Relevant Coursework:* Computer Vision, Topics in Advanced Robotics, High-performance Computing, Optimization

Government Engineering College, Barton Hill, Trivandrum, Bachelor of Technology in Electrical and Electronics Engineering — Aug 2019 – July 2023

- GPA: 8.9/10 (First class with distinction)
- *Additional Coursework:* Minor in Information Technology

Work Experience

Engineer, Backend Development(AI/ML), Qburst Technologies - Trivandrum, India — Aug 2023 – Aug 2025

AI Search Using RAG & Agentic Flow

Tools Used: FastAPI, Langchain, LlamaIndex, MongoDB, Postgres, AzureDevOps.

- The client needed a fast and accurate way to search and analyze their complex global influence data, in which the tables had no proper schema.
- Developed an AI search bar by integrating Retrieval Augmented Generation(RAG) and Natural language to SQL conversion.
- Optimized the system for improved response speed and optimized token usage by more than **56%**.
- Increased the accuracy of search results by **90%** and expanded the range of questions it can handle by **85%**.

AI Report Reviewer System

Tools Used: FastAPI, Langchain, AzureOpenAI.

- The client needed to reduce the high cost and time of reviewing non-uniform diplomatic documents containing text, graphs, and mixed media.
- Developed an AI-powered system for automated evaluation and structured feedback on technical reports.
- Built a core framework leveraging context analysis and coherence checks to ensure unbiased review.
- Designed a custom scoring algorithm to assess document structure, clarity, and relevance.
- Achieved **89%** alignment with human ratings, improved evaluation accuracy by **35%** and reduced review costs by **67%**.

Academic Projects

Multiagent Coordination and Decision Making (RoboCup) — December 2025 – Present

Tools Used: Isaac Sim, MuJoCo, ROS2, Pytorch

- Conducting a multi-agent, real-time learning framework using reinforcement learning and robot learning for the International RoboCup Tournament under the guidance of Prof. Josiah P. Hanna (work in progress).

University Rover Challenge: Obstacle Detection and Avoidance — Sept 2025 – Present

Tools Used: ROS2, Python, DepthAI

- Developed a real-time perception and mapping framework for autonomous navigation on unstructured terrain.
- Implemented a depth-based obstacle detection and reactive control using stereo vision for safe traversal.

Skills & Extracurricular

Programming: Python, C++, Rust, JavaScript, Cuda

AI/ML: PyTorch, Scikit-Learn, LangChain

Tools: FastAPI, Docker, Kubernetes, Git, Linux

Hobbies: Piano, Weightlifting, Badminton, Photography

Perception: Object Detection, Sensor Fusion, SLAM

Robotics: ROS2, Gazebo, DepthAI, MuJoCo, Isaac Sim

Hardware: Raspberry Pi, STM32, Arduino, ESP32

Databases: MongoDB, Postgres, MariaDB, MySQL